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Ultrasonic instrumentation in the food industry

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12.1 Introduction

The commonly accepted definition of ‘ultrasound’ is a pressure wave that is oscillating at a frequency above that of human hearing. Whilst the actual highest frequency audible can vary from person to person, ultrasonic waves can have any frequency above 20 kHz up to a practical limit of around 10 MHz for the applications to be discussed in this chapter. Within this band, the measurement of the speed of propagation, the attenuation, or the reflection of ultrasonic signals can be the basis of instrumentation systems for the food industry. Often these systems have distinct advantages for online food applications, as the ultrasonic sensors normally have no moving parts, can usually be sealed behind stainless steel diaphragms, and occasionally can function as non-contact devices, either above the food product or attached to the outside of the containment vessel or pipe.

Within the ultrasound range, the choice of frequency is dependent on the attenuation characteristics of the medium in which it is to propagate. Table 12.1 gives the traditionally used frequency ranges for process measurements requiring ultrasound transmission. In gases, the attenuation for ultrasound is high, increases as the square of frequency, and is inversely proportional to gas pressure. (See Chapter 13 for an overview on the characteristics of ultrasound propagation in foodstuffs and ambient gases.)

In applications requiring good spatial resolution, short wavelengths are used in preference. For ranging tasks, they are generally in the millimetre range. The corresponding ultrasound frequencies are of the order of 50 kHz for gases and 1 MHz for liquids and solids. Further considerations in the choice of the measurement frequency are the sizes and mechanical properties of the transducer and the cost of the associated instrumentation.

Transducers for ultrasonic equipment are mostly based on piezoelectric materials; most popular are heavy man-made lead zirconate titanate (PZT) ceramic slices with silver-coated top and bottom faces to attach electrical connections. Quartz is a naturally occurring piezoelectric material, but is generally less efficient as a transducer and more expensive. The PZT disks are of various thicknesses to make them resonant at the required frequencies; a 1 MHz crystal is approximately 1.5 mm thick, and a 50 kHz

Table 12.1 Ultrasonic velocity and wavelength examples for typical ultrasound ranging applications

Medium	Usable frequency range	Speed of propagation	Wavelength (at typical frequency)
Gas: air	20 kHz to 100 kHz	332 m/s	6.6 mm (50 kHz)
Liquid: water	200 kHz to 5 kHz	1500 m/s	1.5 mm (1 MHz)
Solid: epoxy	500 kHz to 10 kHz	2500 m/s	2.5 mm (1 MHz)
steel	500 kHz to 10 kHz	6000 m/s	0.6 mm (10 MHz)

crystal 12 mm thick. Most transducers operate in the thickness, or piston, mode of vibration; to achieve a transducer with a directed narrow beam, the disk diameter needs to be around ten times the ultrasonic wavelength used – this means that low-frequency transducers are usually large. The alternative radial mode of vibration of the PZT crystal has been used to produce smaller transducers; for example a 100kHz transducer with a 19mm housing diameter has been described (Lynnworth and Magory 1999).

At the other end of the scale, flaw detection equipment based on ultrasonics is widely used industrially to detect weld defects or cracks in structures. To detect small defects, non-destructive testing (NDT) equipment operates at a very high frequency, and the probes are also required to be very small. A practical limitation in the construction of these transducers is that for frequencies above 5 MHz, the PZT disks are so thin that they become difficult to mount securely against a transducer face.

An industrial or food processing plant is full of sources of ultrasonic noise. All the normal sources of plant vibration or noise also produce inaudible ultrasonic noise. Additionally, there are other sources such as micro-cracks and temperature changes at joints. The ultrasonic transducer-receiver collects these noise inputs, plus electrical noise on the transducer cables in all frequency bands, and then has to pick out the required signals. In general, this is achieved with the self-tuning of the resonant piezoelectric material, receiver-amplifier tuning and microprocessor-based signal recognition techniques. Techniques of coded transmission and correlation detection systems common in radar systems have also been used to detect ultrasonic pulses industrially. The availability of microprocessor power to detect and process these signals has allowed a significant increase in food industry use of ultrasonic instrumentation. The instrumentation will be described separated into low-frequency techniques using airborne ultrasound, and higher frequency techniques where the ultrasound transmission is typically through the process liquid.

12.2 Low-frequency techniques

Instruments using low ultrasonic frequencies, between 10 kHz and 100 kHz, are designed to transmit the ultrasonic beam through air. Such instruments form the basis of two measurement systems relevant to the food industry; pulsed signal echo ranging, and continuous signal echo detection. Echo ranging is used for liquid or solids level measurement, and echo detection provides beam-break ‘presence’ detectors, sometimes used for counting systems.

12.2.1 Level measurement by echo ranging

Instrumentation for ultrasonic level measurement, for example for liquids, is based on pulsed signal transmission followed by detection of the time-delayed echo. A transducer above the surface of the liquid to be monitored transmits a short (approximately 1 ms) burst of ultrasound, typically at 50 kHz. This travels through air (or other gas in the ullage space) vertically down to the liquid surface. Because of the ultrasonic impedance mismatch between the liquid and the air, most of the energy in the airborne beam is reflected vertically back along the same path. By the time the ultrasonic pulse returns to the transducer, it has stopped transmitting and switched over to being a receiver. The time delay between the start of the transmit pulse and detection of the leading edge of the received pulse is measured, and used to calculate the liquid level.

Inherent in this simple description are some major assumptions. The first is that the speed of sound in the air space above the liquid surface is known, or is predictable. The second is that the pulse transmission, reflection from the surface and reception are of high enough signal-to-noise quality that the same part of the pulse is detected in the receiver circuits as is used to trigger the clock on transmission.

Calculating or measuring the speed of sound in the headspace

The speed of sound c in gas is related to pressure P, density ρ and ratio of principal specific heats γ by the expression

$$c = \sqrt{(\gamma P/\rho)} \tag{1}$$

Since for pressures at or near atmospheric values, air pressure is proportional to density, the speed of sound is pressure independent under such conditions:

$$c = \sqrt{(\gamma R)}\sqrt{(T/M)} \tag{2}$$

where T is the absolute temperature and M is the molecular weight. The factor $\sqrt{(\gamma R)}$ includes the ratio of specific heats γ, typically 1.4 for diatomic gases, and the universal gas constant R.

The speed of sound, then, is proportional to the square root of the absolute temperature of an ideal gas. This means that at ambient conditions and 20 ºC the speed of sound increases by 0.17 per cent for every degree of temperature rise, showing that for accurate level measurement, compensation for temperature variation is required. This is most simply achieved by including a temperature sensor in the ultrasonic transducer head and using the microprocessor in the level measurement electronics to compute the appropriate speed of sound.

Equation (2) shows that the speed of sound is not dependent on the pressure of an ideal gas in the ullage space. Over the operating range of ultrasonic systems, and for the gases normally encountered, this relationship is valid and no correction for pressure changes is needed. There are other limitations related to gas pressure. Ultrasound is a pressure wave and it needs gas molecules to interact with each other to propagate through a medium; it cannot travel through a vacuum. In practice, most manufacturers quote a low-pressure limit of -25 kPa (-0.25 bar) gauge pressure, although there is a progressive increase in sound attenuation with decrease in pressure.

The high-pressure limit is set by the transducer itself, that is the stress that its construction can withstand. Since the necessary diaphragm must be flexibly mounted in the transducer housing, the high-pressure limit is typically below 500kPa (5 bar) excess pressure. The actual composition of the gas in the ullage space determines the speed of

sound. Equation (2) shows the relationship to molecular weight, and this is illustrated with data in Table 12.2. Further data can be found from such reference books as Kaye and Laby (1986). In most food industry applications, the ullage space in tanks or vessels will be air with a high water vapour content (humidity). Practical experience has shown that the best accuracy is achieved by setting the *c* value assumed for air to 332 m/s. This dry-air setting is valid up to approximately 40 ºC, but above this temperature, at high humidities, the water vapour (molecular weight 18) increases the speed of sound considerably.

Where such high-humidity, high-temperature conditions prevail, or where high-volatility liquids (solvents or alcohols) are involved, the complex relationship between speed of sound and gas composition makes it impossible for the microprocessors to use data from the pre-programmed tables. In these cases, it is usually necessary to use an ultrasonic technique to measure the speed of sound. In practice, a 'reference reflector' is mounted above the maximum contents level of the tank, and the distance from the transducer to this reflector is measured. The receiver system microprocessor can then compute the operational speed of sound in the variable gas from the time delay of echoes from this known reference. It assumes that this reference path is representative of the whole tank ullage in both vapour composition and temperature, and determines the liquid level using the measured speed of sound. The microprocessor updates sound data at intervals of typically 10 to 30 s.

These alternative techniques are shown in Figs 12.1 and 12.2, with typical transducer constructions. For most food applications, the temperature correction technique is sufficient to allow for the speed of sound variations experienced in practice, and provides a simpler system. The reference pin system will be used only for the highest accuracy requirements or where the variations in ullage gas composition are significant.

Pulse transmission, reflection and reception

The designer of transducers for airborne pulse-echo measurement systems faces the dual challenge: (a) to produce a transmit pulse with a sharp leading edge, so as to allow accurate timing, and (b) to provide the maximum sound energy transmission: all this from

Table 12.2 Speed of sound at º0C in various gases. Values calculated from Kaye and Laby (1986). In this form, they are used to find the start value to be entered. The set value will subsequently be adjusted by the instruments' microprocessor in accordance with the prevailing temperature

Gas or vapour	Speed of sound at 0º C*	Molecular weight
Nitrogen	337	28
Oxygen	315	32
Air (dry)	331.45	29
Methane	435	16
Ammonia	418	17
Water (dry steam)	409	18
Ethylene	316	28
Hydrogen sulphide	296	34
Hydrogen chloride	296	37
Fluorine	283	38
Acetaldehyde	278	44
Freon 11	139	138

* Calculated set value. For gas mixtures, weighted averages are used.

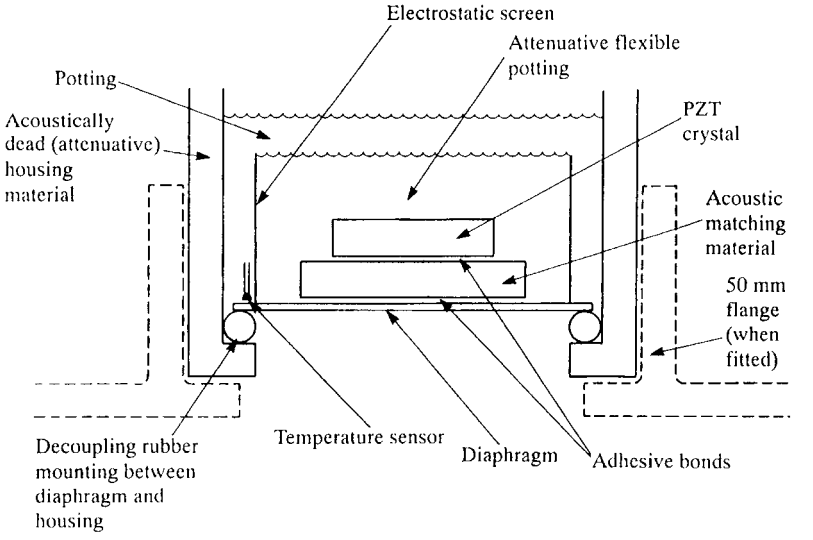


Fig. 12.1 Typical pulse-echo level measurement transducer construction with built-in temperature sensor.

a resonant crystal material, which must then be switched over quickly to work as a receiver. The diaphragm must be of sufficient diameter to produce a narrow beam and flexible enough to vibrate, but must be rigidly attached to the piezoelectric disk. It must then be strong enough to withstand the temperature and pressure ranges required.

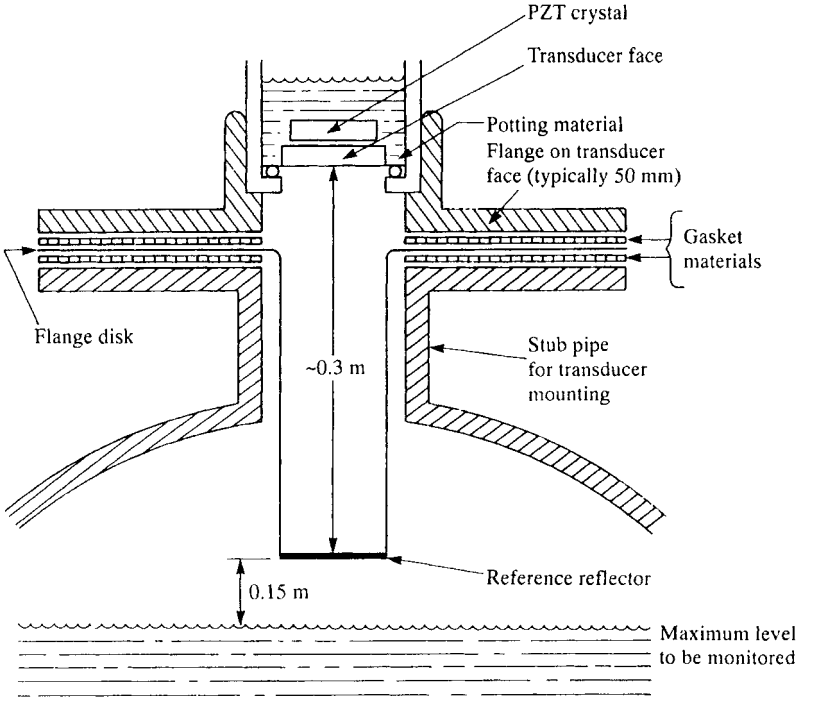


Fig. 12.2 Typical tank installation using reference pin reflector for speed of sound measurement.

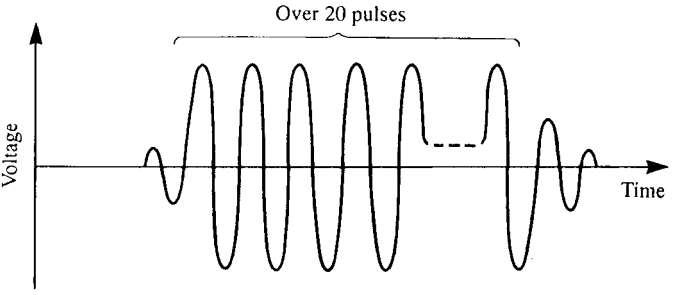


Fig. 12.3 Transmitted pulse train.

In general, a 50 kHz system requires a 50 mm diameter transducer face. This will produce a beam that is within a ±6° angle, but is still economical to fit to process vessels by flanges. For such a system, a twenty-cycle pulse train takes about 0.4 ms to transmit, but the transducer ringing 'after effects' last for a further 1.5 ms. This limits the closest detectable echo to a range of approximately 0.3 m from the transducer face.

The pulse train is now reflected from the liquid surface – from an illuminated area of radius 0.1 m for each metre of range from the transducer. Each ripple or surface discontinuity will give its own reflections, which arrive back at the receiver either with the main echo, or slightly delayed because of the longer path length. Only the first echoes will therefore be recognisable as a pulse train; the later echoes will have secondary reflections superimposed. The waveforms might be as shown in Figs 12.3 and 12.4.

The wavelength between successive peaks of this wave represents approximately 6 mm in distance. It is therefore important for the receiver to identify the correct pulse on the leading edge of the receiver waveform, to remove a potential 6 mm error (or multiples thereof). Using a microprocessor, this is readily achieved. The first technique is to reject echoes that are detected but that fall outside a range gate 'window' set either side of the range of most of the echoes. The second technique is to measure the height of the strongest pulses in the returned echo pulse train, and set a threshold level at, say, 25 per cent of this height. The first echo pulse is then identified as the first detected signal above this automatically adjusted threshold level.

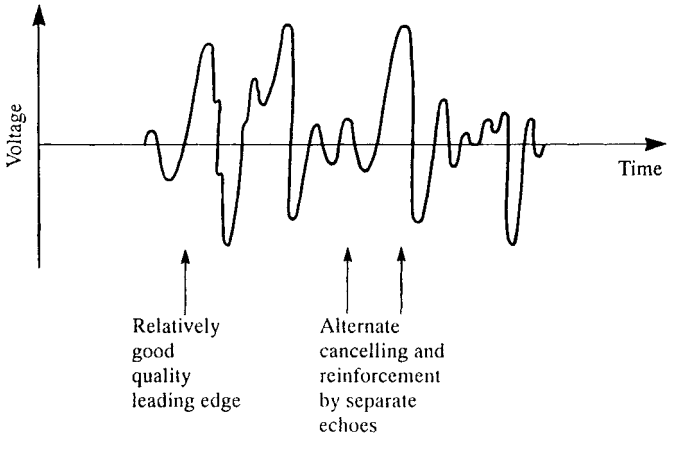


Fig. 12.4 Received pulse train.

Microprocessor-based electronics

The microprocessor is an essential tool in the signal processing and detection circuitry used in pulse-echo level measurement systems. With crystal-based timing circuits, the basic measurements can be made accurately; then the store, hold and compare functions available in the microprocessor allow rejection of invalid echoes on the basis of time delay, signal strength or the fact that they are not confirmed in successive pulses. In addition, the temperature or reference echo detection system can be interrogated on an intermittent basis, to amend the speed of sound value as necessary.

These are the direct measurement functions of the microprocessor. The indirect functions are there as support to improve the system reliability. For example, the cycle time between pulses is subject to a 5 per cent jitter at random around a typical 200 ms cycle for a 10 m range system; this overcomes the effect of any low-frequency cyclical noise. On such a 200 ms cycle, the microprocessor looks for echoes for 60 ms. The remainder of the cycle is used to control the logic of pump or alarm control relays, and set a current output proportional to level, or proportional to tank contents, by using a look-up table. The microprocessor optimises the ultrasonic pulse transmission frequency itself, by adjusting the frequency up or down by, say, 1 kHz. If the surface echo received is stronger at the new frequency, it will track to that frequency for the next 500 cycles. This takes account of temperature effects on the resonant frequency of the PZT crystal itself, which is significant. The typical operational band can vary from 45 to 60 kHz, over the full temperature range of typically −20 ºC to +70 ºC.

The last function of the microprocessor is to decide what to do after a predetermined period of no valid new data. Typically, this can be up to 90 s; after this time the microprocessor can set relays and current output to a safe state, indicate a problem status on the display and even de-energise a fault relay. The most common cause of such a problem is the presence of foam on the liquid surface, which has attenuated the echo to such a degree that it cannot be detected: an understanding of this potential problem is essential to designing a fail-safe measurement system.

Liquids applications

Ultrasonic pulse-echo techniques have been adopted significantly in the food industry since 1983. The first applications were in storage tanks for edible oils. The oil level was monitored with systems using reference pin speed of sound values together with the time-of-flight measurement. Rectangular tanks with approximately 4 m sides were monitored, as well as cylindrical tanks 7 m tall with a conical bottom section. Using the microprocessor to calculate the tank contents in litres allowed blending operations to be monitored and controlled (Solartron Mobrey 1975–91).

A number of dairies have used ultrasonic techniques to monitor the contents of milk tanks, and also of tanks containing aqueous sugar solutions and other liquids on the site. Sugar refineries have used ultrasonics on sugar, molasses and other liquids in storage or process tanks. In all these cases, there is some degree of foaming on the liquid surface, which results in weak echo signals at longer range (for example 5 m). However, the highest degree of foaming occurs during filling, and it is towards the end of this process that a reliable signal is needed to prevent spillage. Because the transducer is top-mounted, the critical point for such high-level alarms is encountered when the liquid surface is around 1 m or less from the transducer. At this range, even the echo from thick foam is usually sufficiently strong to be detectable.

Applications in dairies require that the transducers withstand cleaning processes, even if they do not need to function properly during the cleaning cycle. It is relatively

undemanding for an ultrasound transducer to withstand chemical and hot water cleaning, but steam cleaning can damage, if not blow apart, transducers made from plastic materials. There are now stainless-steel-faced transducers available, but the flexible resins and glues in these will be damaged by anything other than low-pressure steam cleaning (Solartron Mobrey 1975–91).

The use of ultrasonics for monitoring the blending of oils, and successful applications in occasionally foaming sugar and milk, led to applications on yeast tanks. This time, ultrasonic pulse-echo systems were shown to be incapable of reliable operation. In yeast production, the fermentation produces a thick foam. This indeed reflects a sufficient echo signal, but the yeast fermentation produces carbon dioxide gas as well. This carbon dioxide has an unusual characteristic; it attenuates ultrasound severely at the frequencies used in ultrasound echo ranging. (See Section 13.5.2 on the attenuation characteristics of CO_2.) This attenuation prevents the use of ultrasonic pulse-echo techniques in any process that produces carbon dioxide in the ullage space, above concentrations of approximately 5 per cent. This not only applies to yeast fermentation vessels but has also been observed on storage tanks containing blackcurrant and similar juices, which have high carbon dioxide concentrations in the ullage space.

Solids applications

The measurement of solids level in storage tanks suffers from two disadvantages compared with liquids. Firstly, the solids do not flow to achieve a flat surface, so that the returned echoes do not all come from approximately the same range. Secondly, solids level applications tend to be for longer ranges than the liquid equivalents. Large grain silos are a typical example.

The longer range required for grain silo monitoring can be achieved with higher-power and physically larger transducers. It is also useful to reduce the operating frequency, so as to increase the wavelength and therefore reduce scattering by grain or other dust particles in the air. Silos for grain storage have been successfully monitored using such techniques, with reported ranges up to 50 m (Milltronics 1990). The coring effects on such a silo would possibly result in a ±1 m uncertainty as to where the level was, so that even a 1 per cent accurate electronics system giving an error of ±0.5 m at this extreme range would be adequate.

For smaller silos or for solids level measurement applications on conveyor belts or similar positions, the accuracy and resolution resulting from a higher-frequency system would again be needed. Even so, the solid surface level uncertainty is likely to exceed the equipment inaccuracy. In this class, known applications of ultrasonic pulse-echo techniques on solids include monitoring chocolate crumb and peas on tanks feeding process lines, and also monitoring the total thickness of the layer of potato emerging stacked on a conveyor from a slicing machine for potato crisp manufacture. Such measurements of level on a conveyor can be used to give an approximate (total volume per second) flow rate.

Returning to the applications in storage or mixing tanks, ultrasonic non-contact level measurement is widely used for sticky semi-solid substances. Such mixtures occur in the food industry, as pastes and blends, which are then fed to individual containers. The microprocessor can usually be programmed to ignore occasional reflections from the blades or paddles of rotating stirrers, thus giving a true level measurement.

Accuracy

With pulse-echo ultrasonics there are many factors to be taken into account, and a sensibly engineered system will achieve a good result across all of them rather than seek

to perfect the response to one or two. Generally, the microprocessor timing is based on a high-frequency crystal oscillator (e.g. 12 MHz), which is specified across the operating temperature range as better than one part in 20,000, leading to better than 0.01 per cent time measurement accuracy. Timing resolution is typically 1 μs, which represents 0.16mm steps of range in air.

The major source of error is the assumption about the speed of sound in the ullage space, and whether this can be stated more accurately than ±0.2 per cent taking into account variation of ambient gas composition and vapour percentages. Once the composition is established, the temperature averaged over the sound path needs to be estimated. If a temperature sensor is used, how representative is the reading for the sound path average? Inevitably, temperature sensors in transducers cannot track variations in the temperature in the ullage space accurately, and even in steady-state conditions will also monitor the tank surface temperature. This situation can be improved by positioning a temperature sensor separately into the tank, but again the relationship to the average value is questionable. In most cases a further ±1 K error is expected even in steady-state conditions, leading to a ±0.2 per cent error in speed of sound.

The result of these errors is a typical accuracy of ±0.5 per cent in the measurement of liquid surface range from the transducer. With a typical readout showing the tank contents, the range and contents are inversely related; long range means low tank contents. Therefore, as shown in Fig. 12.5, when a 3 m high tank is 10 per cent full, the range error is ±0.5 per cent of 3.1 m (i.e. 1.6 cm). When expressed as a percentage of the depth of the tank contents (0.3 m), this results in a reading inaccuracy calculated as 1.6/30, that is ±5 per cent of the reading. Always, ultrasonic systems of this type have higher uncertainty when tanks are nearly empty, and good accuracy of tank contents when they are nearly full.

In many applications, where for example a certain quantity of liquid has to be dispensed from a storage tank for blending, the repeatability and resolution are more relevant than absolute accuracy of total contents measurement. In such applications, the vapour space over the liquid will remain constant in its temperature and composition, making the dispensed quantity itself measurable to an accuracy of ±0.5 per cent. With the resolution of the timing measurement typically equivalent to ±0.2 mm of liquid level change, the main limitation will be the method of reading the level change. The digital display can usually give the highest resolution to an operator, typically better than that supplied by the current output from the measurement electronics.

The simplest microprocessor systems with 8-bit mathematics had 256 steps between empty and full, i.e. across the range of the current output. The smallest step change transmitted was therefore 0.4 per cent of the span of this current output. A similar 256-step look-up table was used to relate liquid level to tank contents, where the microprocessor is used to derive tank volume; this used to result in approximately a 0.4 per cent (of volume) minimum step size on the electronics display. Microprocessor systems using 12-bit mathematics result in 0.025% resolution, that is 4000 steps in these look-up tables and the current output.

Short-range transmitters

The development of relatively short-range pulse echo level measurement systems has recently bridged the gap between proximity detection sensors and tank level monitors. These are available as complete potted transmitters, with ranges from centimetres to metres. (Pepperl and Fuchs 2000; Nivelco 2000). Inevitably these will be used for control of fill systems for small liquid containers. A similar food application was in the detection

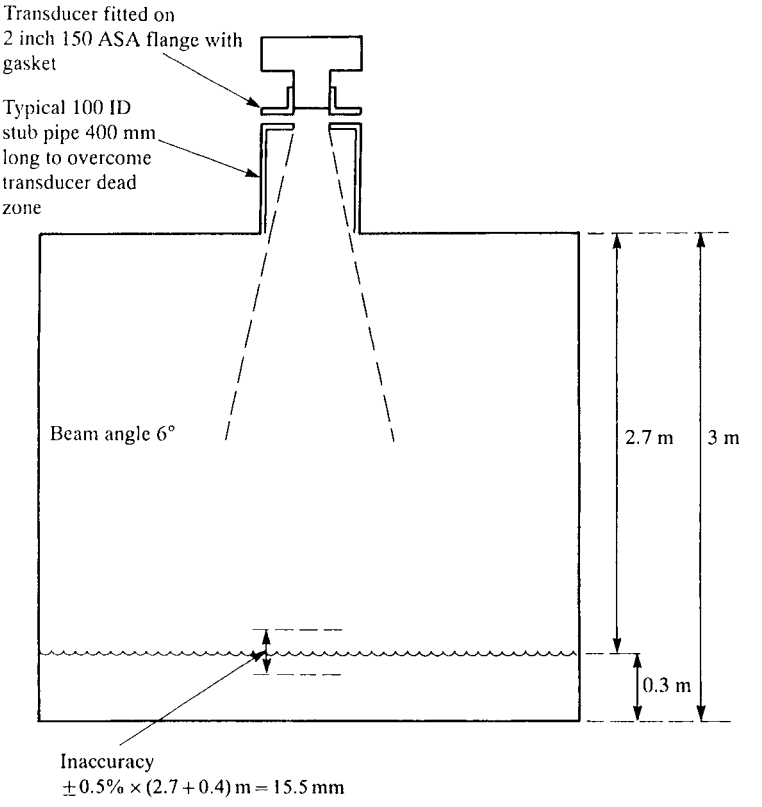


Fig. 12.5 Ultrasonic system level measurement installation and accuracy example.

of a double fill of dough being loaded into a baking tray in error, which caused jamming of the automated baking process (Siemens 2000). Another unusual application has been for density measurement, where a float suspended in a solution of alcohol/water, in a tank above a weir, floats at a height dependent on the solution specific gravity (SG). The float position is monitored using a short-range ultrasonic position transmitter (Ixthus 1999) which can be used as an SG measurement.

12.2.2 Beam-break detectors as counters

Since any relatively solid object placed in an ultrasonic beam will reflect or deflect the sound wave, systems have been developed to detect the presence of such solid objects using the interruption of an ultrasonic beam between transmitter and receiver. Working at 30–50 kHz, such systems are used as counters for packages on conveyor belts.

The advantage of ultrasonics over other techniques such as optical or infra-red beam-break detection systems is that the ultrasonic beam is less severely attenuated by dust in the air, and more importantly is unaffected by dust build-up on the transducer faces themselves. This advantage arises because the transducer faces physically move to transmit the beam, rather than being purely a window as in the optical system. Ultrasonic beam-break detection techniques are also used as set point level controls (employing transducers mounted on opposite sides of the tank), particularly on dusty materials such as grain in silos.

12.2.3 Future developments in low-frequency instrumentation

The main developments in low-frequency ultrasonic pulse-echo techniques will be in improvements to the design of transducers. Already existing designs are capable of operating at temperatures of 130 ºC and pressures of 500 kPa (5 bar) gauge; and are available in different materials of construction such as plastics, PTFE and stainless steel. With existing styles of PZT transducers these limits are not likely to be significantly increased. New designs of transducer are becoming available, where the piezoelectric is separated from the temperature extreme, by interposing an acoustic waveguide or buffer. This technique will also allow operation at higher pressures (Lynnworth and Magory 1999). In performance, the only real change anticipated will be in the closest allowable measurement distance, which might be reduced below 0.1 m, for a 10 metre transducer, probably by using radar type signal coding and processing techniques.

The development of a clamp-on liquid level transmitter for use attached to the outside of a tank wall, using these low-frequency techniques, is unlikely because of the acoustic mismatch involved. However, clamp-on ultrasonic sensors for gas flow measurement have been described operating on plastic and even steel pipes (Brown and Lynnworth 2001). Higher-frequency clamp-on techniques for level measurement are described in Section 12.3.3. Techniques have been developed, notably by UKAEA, Harwell, for using a tube to guide the airborne ultrasonic pulse down into a tank (Asher 1983). This could be of use in the food industry to avoid beam interference from major tank obstructions such as heating coils or multiple stirrer systems.

As all industrial instrumentation has moved towards simple transmitters and operation on fieldbus systems, ultrasonic level sensors are following the trend. Sensors are already available as 4-20mA transmitters, with intelligent digital communications able to transmit information over a fieldbus link, as well as re-programme the operation parameters of the unit.

12.3 High-frequency techniques

Instruments using higher ultrasonic frequencies, between 500 kHz and 5 MHz, are designed to transmit the beam through the liquid, and in some cases through the tank or pipe wall to reach the liquid to be measured. Systems in this frequency range are based on the measurement of one of the following:

- the attenuation of the beam (measuring the amplitude at a receiver)
- the propagation velocity of the beam in the liquid (measuring the delay of a pulse)
- the frequency of a returned received signal
- the loading of the liquid on a transmitted signal (monitoring the decay or reflection of an ultrasonic pulse resulting from the impedance mismatch between transducer surface and liquid).

Based on these techniques, liquid level, liquid flow, suspended solids and liquid composition monitoring systems applicable to the food industry have been commercially developed.

12.3.1 Flow-measurement systems

Ultrasonic flow-measurement systems have the major application advantage that, in most cases, the transducers can be mounted in or on the walls of the pipe carrying the flow to

be monitored. They are therefore obstructionless, which is a major advantage, but can also be non-contact devices, and thus have the further major advantage of being hygienic.

In medicine, blood flow measurement pioneered the use of ultrasonic flow meters based on the Doppler effect. These were then applied industrially from 1977 onwards, with many applications in the food industry. Work at Harwell led to the time-of-flight ultrasonic flow measurement system at about the same time, but the two techniques are totally different in their underlying principles and performance characteristics, and lead to different applications. Recently the time of flight technique has been developed into a clamp-on device, which is very useful in food applications.

Doppler flow measurement

The Doppler effect is characterised by a change in the frequency of a wave produced by or reflected by an object that is moving towards or away from the observer. In a Doppler ultrasonic flow meter, a piezoelectric crystal transducer produces an ultrasonic beam, which is launched into a flowing liquid. Discontinuities in that liquid, typically suspended solids, reflect the beam, and if these solids are moving towards the transducer then the reflected beam has a higher frequency. The frequency shift, as a percentage of the transmitted frequency, is proportional to the speed of movement of the suspended solids, as a percentage of the speed of sound in the liquid.

The transducer is normally clamped on to the outside of the pipeline, using silicone grease or epoxy resin as an acoustic couplant between the transducer face and the pipe wall (Fig. 12.6). Within the transducer housing there are usually two piezoelectric crystals, angled by typically 30° from perpendicular to the pipe surface. One crystal transmits a 1 MHz continuous wave signal into the pipe and the flowing liquid. The returned reflected signals are detected by the receiver crystal and mixed with direct coupled transmitted signals. Various electronic circuit detection systems are used to average the returned frequency shifted signal; to obtain a characteristic frequency quoted as the Doppler frequency. This is then processed to give flow rate.

Because of their advantages – clamp-on, obstructionless and hygienic, Doppler flow meters are widely used in the food industry, particularly on liquids containing particulates, for example soups or jams. Slurries also present excellent applications; for

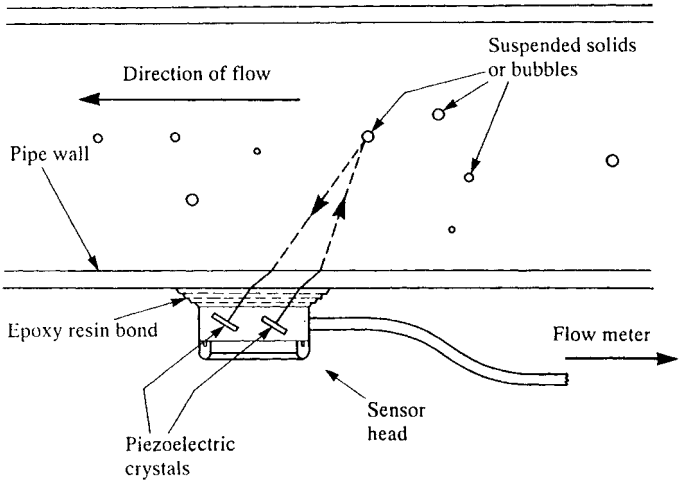


Fig. 12.6 Doppler flow meter sensor.

example, the sugar beet slurry produced in the first stages of the refining process is measured with a Doppler flow meter. In brewing, Doppler flow switches are used on the pipelines containing wort and yeast, and in sugar refining the flow control systems on lime milk are based on Doppler flow meters (Solartron Mobrey 1975–91). Over the 25 years that Doppler systems have been sold, the designs have improved and transmitters are now available: these modern systems are being used on food applications like salmon paste slurry and orange juice (Pulsar 2000).

While the Doppler technique is ideal for slurries, or any liquid containing solid particles, its advantages of clamp-on configuration affording convenience and hygiene make it very attractive for other flow measurement duties, sometimes on clean liquids. Results are often reported to be good despite the lack of apparent reflectors. Whilst this is often ascribed to reflections from turbulence or eddies, the reflected signal is more likely to come from entrained air bubbles. Such techniques have been used for flow failure alarms, siting Doppler type sensors near pumps to indicate when the water flow has ceased and the pump is running dry. It is very important to use care in the application of Doppler techniques to flow measurement applications in clean liquids. The normal plant noise and vibration of the transducer on the pipe wall can feed an apparent 'Doppler' frequency signal into the detection circuits, giving a false flow measurement.

To increase confidence in the Doppler signal, most Doppler flow meters have a signal strength meter to measure the amplitude of the returned signal. This can be used to give a measure of the reliability of the flow reading. On relatively clear liquids there are many applications where good results can be obtained. Milk and orange or other fruit juices normally give good results. Some liquids such as oils and condensate in return lines give poor results, but such is the enthusiasm for the use of Doppler systems that at times air bubble injection systems have been used to improve the flow meter performance on these liquids. This is perhaps acceptable in plant commissioning or proving trials, where a portable Doppler flow meter might be used for troubleshooting, but is hardly recommended for permanent installation.

With most Doppler systems being clamped to the pipe wall, it is important to consider the effect of the pipe wall as part of the transducer. The inside surface must be smooth and parallel to the outside surface, and made of a single continuous material. This excludes lined pipes, where any gap between the lining and the pipe material will stop all transmission. It also excludes inhomogeneous materials like cement, cast iron or glass reinforced plastic (GRP). Any paint on the outside of the pipe should preferably be removed. The best types of pipe are plastic or glass, because they transmit the ultrasound efficiently through to the liquid and yet do not allow pipe noise or plant vibration to travel too far. While the speed of sound in the liquid will affect the angle of refraction of the ultrasonic beam, and this therefore varies with temperature and type of liquid, the frequency shift is dependent on the component of the liquid velocity along this beam angle. A mathematical evaluation of beam refraction theory shows that the Doppler shift is independent of liquid temperature and composition, provided that the pipe walls are parallel to one another and the pipe axis (Cousins 1978; Scott 1979).

Doppler flow measurement systems are most useful on pipe sizes from 20 to 200 mm, monitoring flows where the linear velocity along the pipe is normally above 0.3 m/s. In such cases the accuracy achieved will be better than 5 per cent, although many users will claim much better results than this. The main application is in flow measurement of slurries, or any liquid containing a second phase in suspension. For liquids with a heavy suspended solids loading or a high density of entrained gas bubbles, the resulting strong

attenuation of the ultrasound signal means that the flow velocity is measured close to the pipe wall only. This leads to an error in the derived value for the pipe overall velocity.

Time-of-flight flow measurement

The original design of this technique used two ultrasonic transducers fitted into bosses welded on to a section of pipe. They are positioned so that an ultrasonic beam transmitted from one to the other travels at an angle of typically 60º to the pipe axis and therefore the liquid flow (Fig. 12.7). A pulse of ultrasound is timed travelling through the liquid between transducer 1 and transducer 2; because the pulse is travelling downstream with the flow, the transit time is shorter than that for a similar pulse transmitted upstream from transducer 2 to transducer 1. Electronics systems driving the transducers can be used to derive the difference between these two transit times. Typical values for a 100 mm-bore pipe might be a transit time of 100 μs, and a difference in transit times of 100 ns for a 1m/s flow velocity. Measuring this to 1 per cent accuracy requires a stability of 1 ns in this time measurement, which is easily achievable.

Where the flowing liquid contains suspended solids, these solids will scatter the sound and attenuate the pulse as it travels between transducers. This will prevent reliable operation on liquids where the solids loading is too great. Temperature changes or dissolved solids that affect the speed of sound are automatically compensated for in the electronics, by measurement of the average transit time between transducers to derive the actual speed of sound in the liquid. The major application for this technique is in the larger diameter pipeline applications of the water supply industry. Here the ability to install transducers into bosses welded onto existing pipe-work, even without interrupting the flow, is a major advantage. Systems are installed on pipelines of up to 3 m diameter, monitoring flow with accuracies better than 0.5 per cent of reading. Such systems have been installed to measure wastewater flows in sewers, and on site for process control in both sewage treatment works and water purification plants (Solartron Mobrey 1975–91).

Flow measurement applications in other parts of the food industry are less numerous, since the advantages of this technique are less significant there, and difficulties in installation occur in smaller-diameter pipelines. To provide sufficient ultrasonic path length for pipe diameters below 75 mm, the transducers are usually positioned to send the ultrasonic beam axially along part of the pipe, achieved by introducing a double bend in the pipe-work. Despite the flow profile distortion, very good linearity has been achieved on examples of such systems.

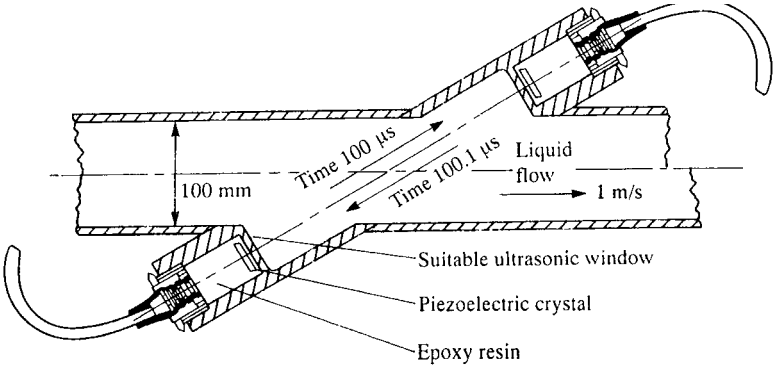


Fig. 12.7 Time-of-flight flow-measurement system.

Time-of-flight flow measurement using clamp-on sensors

The time-of flight technique has been significantly developed in recent years, and the range of applicable pipe diameters extended down to 1 inch and smaller, particularly using transducers externally clamped to the walls of the process pipe-work. Variations exist with certain models having sensors on opposite sides of the pipe, and other models relying on reflection of the beam from the far wall, and having both sensors on the same side of the pipe. Most success has been with plastic pipes, because the transmission of sound through the wall at an angle is easier in these materials, where the acoustic impedance is closest to that of the liquid to be measured.

In creating a flowmeter using the pipe wall as part of the acoustic path, it is necessary to know the wall thickness and material properties, to define the transducer spacing needed, and to calculate the volume flow for a measured velocity. Many flowmeters overcome this by using the operator to position the sensors at the separation giving maximum signal strength, and even by having separate circuits to monitor the wall thickness ultrasonically. But at present many of the clamp-on systems are dependent on operator skill and experience to achieve useful results. Some systems require separate measurement or calibration for different speeds of sound in the liquid being monitored, which can change depending on the process.

Potential use of these techniques in food is enormous, but real applications will be for single point non-portable devices, at a point in the process where the flowmeter has been checked and found to work reliably. Currently there are systems available from many suppliers, (Panametrics 2000; Micronics 2000; Flotec/Valeport 1999; Katronic/Flexim 2000), who claim various food applications. A Japanese application in a dairy has clamp-on meters used to monitor the cleaning fluid velocity, which increases once the pipes are clean, and so signals the end of the cleaning cycle. Pipes were SS304 or 316 sanitary material, 1.5 to 4 inches, internal diameter (ID) (Panametrics 2000). A special product design for this application has been introduced for CIP cleaning process control in the UK (Able 2000). Another application was for fruit juice measurement, both concentrate and diluted, water and condensate, on sanitary stainless steel tubing 1.5 to 2.5 inches ID, 0.065 inch wall (Panametrics 2000). Recent tests reported were on custard and chocolate lines, typically 2 inches in stainless steel and plastic pipe materials (Katronic 2000).

12.3.2 Liquid level switches

Liquid level sensors based on ultrasonic transmission have been manufactured commercially since 1973, and have evolved into various styles of sensor for high- or low-level alarm, either penetrating the tank or for use across a vessel or pipe as non-penetration systems. Advances in transducers and electronics have recently made pulse attenuation measurement systems viable, which has led to a device for liquid level control without penetration of the vessel wall.

A series of level switches can be chosen as an alternative to the echo ranging systems for level measurement (Section 12.2.1). Level switches (point sensors) offer the following advantages over the ultrasonic level measurement techniques: steam cleanability, high-pressure tolerance, fail-safe features, reliability, low cost, and insensitivity to ullage gas composition and to stirring paddles.

Ultrasonic-transmission-based level detectors

At an operational frequency of 1 MHz and above, the attenuation of ultrasound in air is very high. Transmission through liquids, on the other hand, is very good, so that when

two transducers face one another, ultrasound will only pass from transmitter to receiver when the gap between them is filled with liquid. The technique is used in a liquid level sensor circuit as shown in Figure 12.8 (Solartron Mobrey 1975–91).

In practice, the transmitter crystal is driven from the output of a tuned RF amplifier, with the receiver linked directly to the input. The system sensitivity is adjusted using the gain potentiometer on the amplifier. With liquid in the sensor gap, the loop gain is greater than unity, and the system oscillates at its resonant frequency. This is detected on the output of the amplifier and an output relay is energised. When air only is present in the gap, the coupling between the amplifier output and the input is from the cross-talk between the two coaxial cables to the sensor, and from ultrasonic energy that travels from the transmitter crystal down the 'bridge' to the receiver crystal. Direct coupling through the air is negligible.

All the components in this feedback loop have resonant frequencies that can vary with temperature and liquid type. The stainless steel windows of the transducer itself must be machined to the correct thickness; but more surprising is the effect of the gap between the two transducer faces filled with liquid. Because of the mismatch between the acoustic impedances of steel and water, or any other liquid, standing waves are set up of multiple reflections backwards and forwards in this gap. By using a self-tuning self-seeking resonant feedback circuit, the oscillation settles on the frequency that gives the highest loop gain, and tracks this with temperature and variations of fluid composition which affect the speed of sound.

Different designs of sensor can be produced, based on this same principle, using bodies made from different materials, for example plastic or epoxy resins or PTFE. There are stainless steel sensors of different designs, where the 'bridge' is less significant in

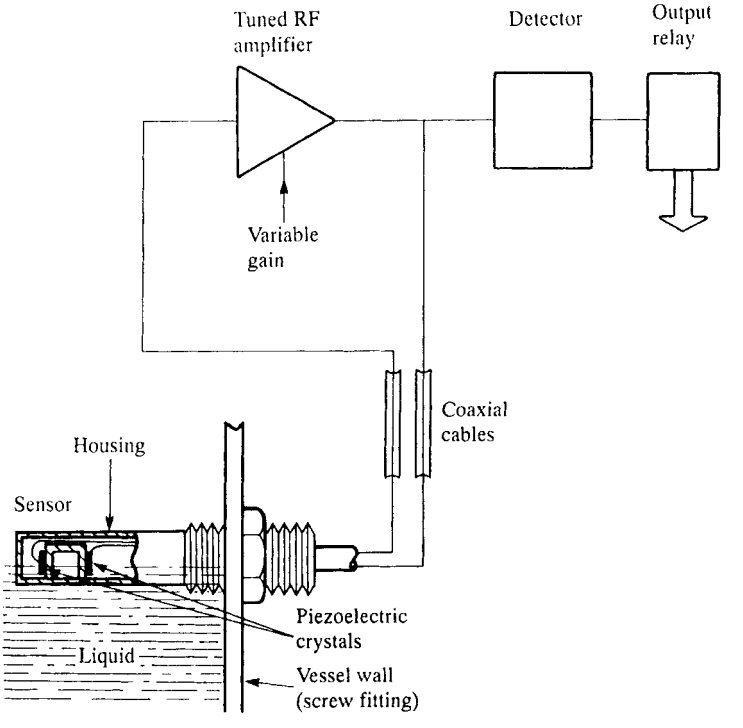


Fig. 12.8 Ultrasonic liquid level sensor circuit.

producing false dry signals. The main advantages for the user are that the sensor is small, has no crevices or moving parts and is suitable for both conductive and non-conductive liquids. It ignores foam and moisture droplets and is unaffected by most coatings. The disadvantages reside in the relatively low maximum temperature for the standard unit (150 °C) and the effects of bubbles or suspended solids. Either of these attenuates the signal passing between transmitter and receiver because the second phase scatters the ultrasound. To overcome this it is necessary to increase the gain of the amplifier, and usually a different sensor design is used for such liquids where the 'bridge' is less significant, producing less coupling, so that the amplifier gain can safely be increased to a high level. Special sensors have therefore been developed specifically for aerated or carbonated liquids and for liquids containing suspended solids (Fig. 12.9), using extended bridges. The use of epoxy resin or plastic sensor housings improves the acoustic match between sensor and liquid, therefore increasing the signal obtained in a liquid.

Applications for this type of sensor in the food industry abound, both for the standard sensor designs and for designs based on these principles for specific applications. General-purpose high-level, low-level and pump control duties are standard applications in all types of liquid or beverage. Aerated liquids, or those with suspended solids, require a sensor of a slightly different design, to allow higher gain in the amplifier unit, overcoming the extra attenuation from these liquids; this is discussed later. The small size of the sensors is an advantage when detection is required in a small space, such as in the discharge lines from blending tanks, to ensure that the tank has been emptied. A specific example of this was in a whisky-blending plant, where the requirement was that the sensor should not treat whisky foam the same as pure liquid. In fact, this type of sensor treats foam the same as air. HM Customs and Excise required the control system to indicate tank empty, but did not wish to insist that all the foam had been drawn off too. Similarly, sensors are used in delivery pipelines to alarm when the pipe is no longer full, when a flow meter reading will no longer be valid or when a coolant flow has failed.

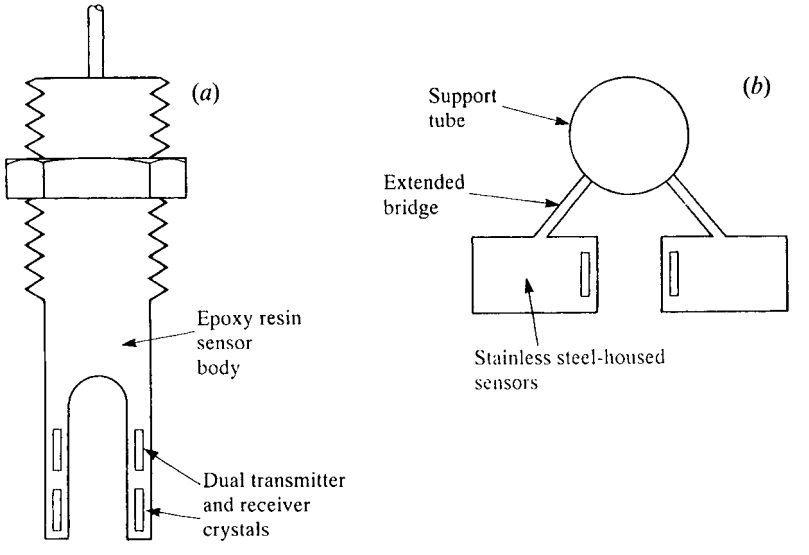


Fig. 12.9 Special sensor designs: (a) sensor for aerated liquids; (b) top view of sensor for liquids containing suspended solids.

Specific sensor developments have been made for cryogenic applications, to control the level of liquid nitrogen, for example, in vessels for freezing fresh fruit such as raspberries. For beer-brewing systems the insensitivity to foam has been used to advantage, with a sensor on the end of a tape to allow hand dipping of the tanks. This replaced the standard HM Customs and Excise approach, using a dipstick, which is relatively difficult to apply in certain real ales. Another special sensor was developed for one customer based on a section of pipe, with the piezoelectric crystals potted into recesses machined in the outside of the pipe walls. The inside of the pipe was therefore of normal smooth bore, which achieved the very high standard of hygiene required on this plant. Transmitting through a pipe wall from the outside is relatively easy, although limited to pipe diameters between 20 and 75 mm. The standard approach uses two separate epoxy resin transmitter and receiver blocks. These are bonded to the outside of the pipe wall, while held accurately parallel. This technique is regularly used on sight glasses attached to beer vats, to give high- and low-level alarms for automatic control. It is particularly useful when the level is no longer visible to an operator because of discoloration of the sight glasses.

A development of this ultrasonic technique has been to turn the sensor 'inside out' by mounting the piezoelectric crystals on the inside of a cylinder (Fig. 12.10). The ultrasonic beam travels around the cylinder as a plate or Lamb wave (see Section 13.3.2), but can be thought of as reflecting backwards and forwards within the thick wall of the transducer. At each outside surface reflection, where the sensor is immersed in liquid, a proportion of the energy is transmitted into the liquid. This is lost in the surrounding volume, and the wave is attenuated compared with the direct coupling between the two transducers when the sensor is in air. Since this transducer is active when it is dry, it is suitable for a fail-safe high-level alarm, and this is its main duty and leads to the name HiSens. In the food industry, the main applications are in overfilling alarm systems on process vessels or cooking vessels (Solartron Mobrey 1975–91).

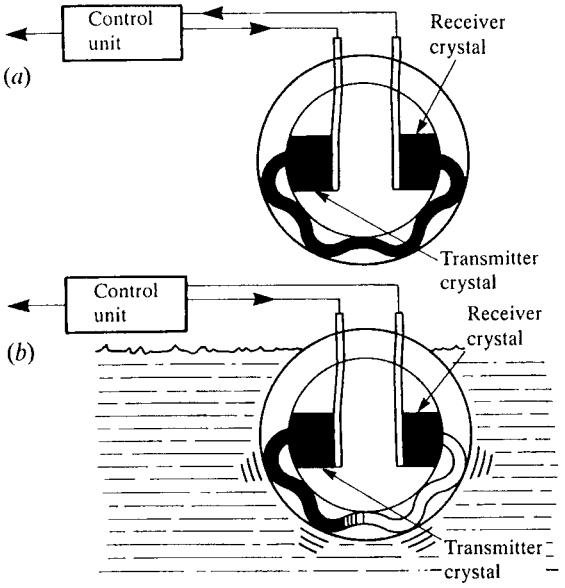


Fig. 12.10 Mobrey Hi-Sens sensor. (a) Sensor in air: ultrasonic signal reaches crystal. (b) Sensor in liquid: ultrasonic signal is lost into the liquid and does not reach receiver crystal.

A major limitation of the ultrasonic technique must be mentioned, which particularly relates to gap-type sensors. Because these sensors rely on transmitting the ultrasonic signal between the transducers, through the liquid, the assumption is that the liquid will transmit this sound relatively easily. Special sensors have been designed for use in liquids with suspended solids or for carbonated liquids. A more severe problem occurs in the dairy industry with milk containing a high proportion of cream. Either because this cream produces an emulsion layer, where the second-phase fat droplets act as scattering centres for the ultrasound beam, or because the highly viscous cream layers trap air bubbles, following agitation or splashing of the milk, the ultrasonic beam is highly attenuated. In many cases ultrasonic gap sensors have been found unsuitable for level detection in such dairy applications. The possibility of using this effect to detect the difference between cream layers and milk is a form of interface detection (see Section 12.3.4).

Pulse-attenuation-based level detectors

The old school of practical engineers will tap the outside of a tank and listen for a different 'dull' sound when the tapping is below the unseen liquid level. This technique is taken one step further using an ultrasonic transducer and 'near wall attenuation' measurements as originally proposed by UKAEA Harwell (Asher 1983). A high-frequency (2 MHz) pulse is coupled into the vessel wall from an externally bonded transducer. If the internal face of the wall is under the liquid, some of this pulse is transmitted into the liquid and lost in the volume of the vessel. If the internal face is in air, the ultrasound pulse is all reflected back towards the transducer. The proportion of the ultrasound reflected from this boundary is dependent on the acoustic impedance mismatch between the vessel wall and the liquid or air behind the wall. For any gas, the acoustic impedance (which is the product of sound velocity and density) is so low that the reflections within the metal wall are total. For a liquid, there is still a considerable mismatch, but the proportion reflected is lower than 100 per cent, and after many traverses of the vessel wall, the ultrasonic pulse amplitude is significantly reduced (Fig. 12.11). By adjustment of the signal amplitude threshold for detection and the time delay before a measurement is taken, a non-penetration liquid presence detection system can be produced.

The advantages of this technique are that it is hygienic, it requires access to one outside surface of the vessel only, and it can be retrofitted by external clamping or

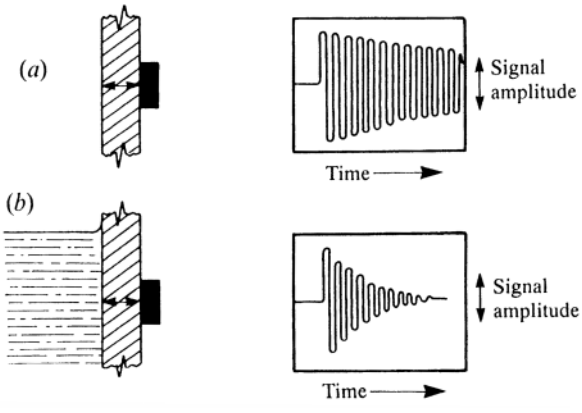


Fig. 12.11 Near wall attenuation level sensor: (a) vessel empty, (b) vessel full.

bonding to existing tanks or pipe-work. A significant application in the food industry has been in fitting level control sensors to refrigerant liquid tanks on commercial cold cabinets and freezer installations. The disadvantages are those of on-site adjustment and set-up procedures which are an inevitable part of any system that involves the vessel wall as part of a transducer system. Recent product introductions based on this principle have been used on milk tanks and in the brewing industry, and appear to have simplified the set-up procedure (Pulsar 2000).

Pulse-transit-time based level detectors

This technique uses a pulse echo system, where a single transducer mounted on the outside of a tank transmits a pulse of ultrasound into the tank wall. If the inside of the wall at that point is below the liquid surface, some of this sound energy will be transmitted into the liquid. The beam of ultrasound traverses the tank, is reflected off the far wall, and returns to the same sensor, provided the tank geometry is correct and there are no obstructions like stirrers. The receiver circuitry detects this time-delayed pulse. Detection of the pulse at approximately the right time delay from transmission indicates the presence of liquid in the tank at that level. If there is no liquid present, then there will be no returned echo, because the ultrasound cannot pass through the air without significant extra attenuation, and any energy 'ringing' in the wall itself should have died away earlier. Further applications of this technique are given below.

A different style of transit time sensor uses two transducers clamped to the outside of the tank wall. These are the same style of sensor used for liquid flow measurement, and the same flowmeter electronics can be adapted. A flexural wave is transmitted along the short (approx 225 mm) horizontal path between the sensors, and the velocity of this wave is reduced when liquid is present inside the tank at that level (Lynnworth and Magori 1999).

12.3.3 Liquid level measurement systems

Pulse echo techniques 'up from the bottom' have been developed particularly for application in the brewing industry. The investigation of liquid level measurement using high-frequency ultrasound transmitted in the vessel wall (or equivalent) has produced many prototype techniques based on research but not many practical or commercial systems.

Pulse echo techniques in the liquid

Ultrasonic pulse-echo techniques using a transducer either in the liquid or bonded to the tank base have been used successfully in the brewing industry in particular. In this system, the ultrasonic pulse from the bottom-mounted transducer is timed up to the liquid surface and back through the liquid (Fig. 12.12). Because it is necessary to know the speed of sound in the liquid, a separately mounted transducer on the side of the vessel is used to measure the time delay across the vessel in a horizontal plane – a known distance. This will correct for the variations that will result from temperature and liquid composition changes. The microprocessor-based electronics then compute the required level in the tank, which can be expressed as volume using the known tank shape. This technique does not suffer from the surface foam problem that severely restricts airborne pulse-echo systems in this industry. Using a liquid as the transmission medium, the transducer is more efficient. The signal attenuation in a clear liquid is low, but becomes higher when any gas-producing fermentation occurs or when any solids are in suspension,

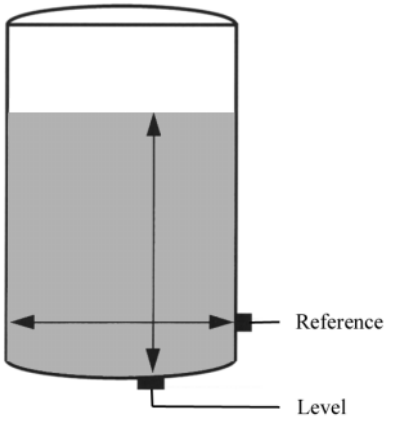


Fig. 12.12 Pulse echo level measurement using a bottom mounted transducer. (Courtesy Canongate Technology).

because of the scattering produced by these second phases. In tanks where the sediment build-up is considerable, the transducers can be positioned off the bottom of the tank. The technique has been used widely in breweries, but has also been applied to cider, wine and spirit tanks. Usually in a multi-tank installation the electronics is multiplexed between separate transducers on up to sixteen tanks (Canongate 2000).

Other pulse echo techniques

By monitoring the attenuation or reflection of plate (Lamb) waves (see Section 13.3.2) in a metallic vessel wall or probe inserted into a vessel, the change occurring when this surface is submerged in a liquid can be measured. Analogue or time-based measurements can be used to deduce a liquid level. The technique has been used to monitor ultrasound pulses in a vertical rod inside the vessel, penetrating into the liquid. System accuracy is enhanced by having a magnet in a float on this rod and using the ultrasound pulse to detect the magnetic discontinuity produced at the liquid surface level; this is really a float level measurement that uses an ultrasonic position sensor (Eurogauge 2000). Such techniques have been limited to measurements in the nuclear industry or for high-specification/high-accuracy applications.

12.3.4 Suspended solids and interface detection systems

The technique previously described for liquid level switches – the monitoring of ultrasonic transmission amplitude between transmitter and receiver crystals – can be used to detect different characteristics of the liquids being monitored. The most significant measurement of this type is of the concentration of suspended solids, which increase ultrasound attenuation because they act as scattering centres. Similar systems have also been used to detect interfaces between clear liquids, by use of either the different attenuation in each liquid or the reflections that occur at the interface. An emulsion contains two phases separated by many interfaces. The increased attenuation due to scattering provides a means of distinguishing the emulsion from the single-phase liquid without second-phase inclusions.

A sensor used as an ultrasonic attenuation monitor is typically either suspended at a fixed point in a tank or used to monitor a liquid flowing down a pipeline. For either case,

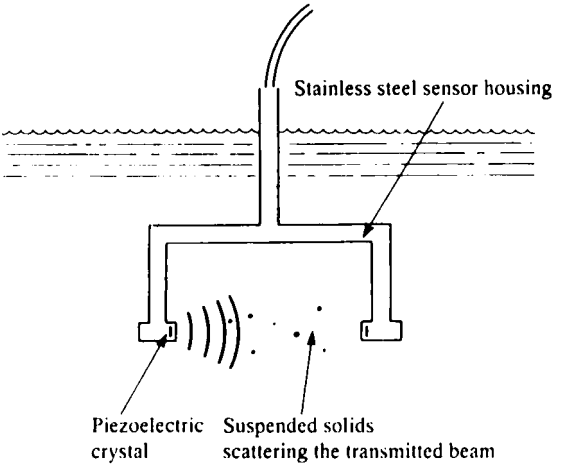


Fig. 12.13 Sensor for blanket detection in sedimentation tanks.

a gap between the sensor faces of between 100 and 300 mm is typical, to provide an adequate liquid sample length. For a suspended sensor in a settlement tank, a 150 mm gap device would be used, constructed as a fork arrangement (Fig. 12.13). For pipeline use, the sensors would be mounted in bosses welded onto the side of the pipe, and would have their faces flush with the inside wall to minimise flow obstruction and silting up of the sensors (Fig. 12.14).

Suspended solids measurement

In practice, it is found that the attenuation produced by suspended solids particles, when expressed in decibels (dB), is proportional to:

- the gap between the sensors over the range 20–600 mm

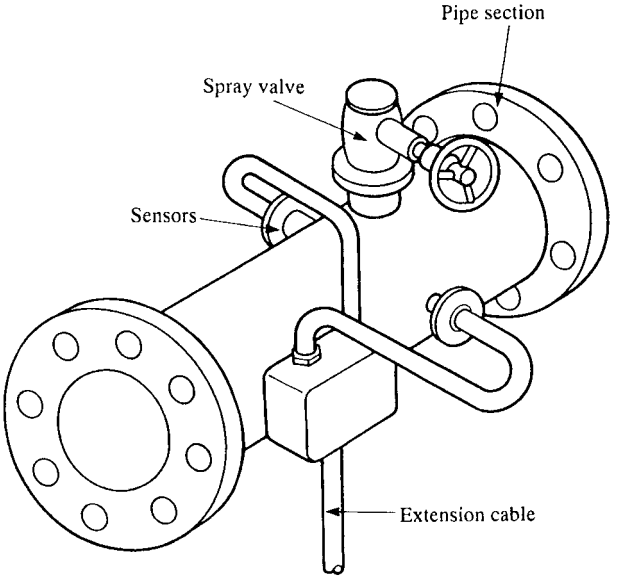


Fig. 12.14 Suspended solids measurement using pipe-mounted sensors.

- the ultrasonic frequency over the range 1–4 MHz (wavelengths approximately 0.5–2 mm)
- the percentage of solids by weight in the solid-liquid mixture, over the range 0.5–50 per cent.

The proportionality of ultrasonic attenuation to weight percentage of solids applies to solids in suspension that have submillimetre particle sizes, i.e. radii smaller than the wavelength. This attenuation is a result of ultrasonic scattering rather than beam interruption or diffraction, which occurs when the wavelength is smaller than the particle size (Flammer 1962; Lenn 1987).

Different solids in suspension have different characteristics of attenuation, which relate to the size and density of the solid particles and the acoustic properties of the solid material. In general, practical experimentation is necessary to establish these characteristics to choose the correct combination of frequency and sensor gap so that the monitoring electronics give an output of percentage solids over the required range. There are two basic control output requirements for suspended solids monitors, either a relay high/low output or an analogue output proportional to suspended solids. In the simpler relay output system, a gain control is used to set a fixed attenuation level, which defines the transition from high to low solids concentration. This is used in settling tanks typically as a blanket level alarm, to trigger or stop the desludging of a tank.

In the water industry, such ultrasonic sensors are used for water treatment floc blanket alarms, to prevent the floc from spilling over into the water supply system. Similarly, they are used for control of sand filter backwash systems, alarming when the bed is fluidised too much and in danger of being washed away. Similar techniques have been investigated by the Brewing Research Foundation for use in brewery fluidised bed backwash systems. More severe sludges are met in the sewage industry, where primary sedimentation tank control systems are normally based on ultrasonic attenuation sensors. Many food-processing sludges can be treated in a similar way. For example, Australian sugar cane processing uses ultrasonic blanket detectors in the subsiders, to separate juice and suspended impurities.

The analogue output system for measuring the percentage solids of a slurry has been used primarily on mineral slurries such as china clay and coal tailings, but has also been used in the food industry. In the UK sugar beet industry, lime additions are monitored using these techniques. In the brewing industry, recirculation of the yeast in the sediments via membrane filters is controlled based on such measurements. Other applications have been found in peanut processing in the USA (Solartron Mobrey 1975–91).

The electronics used to monitor the ultrasonic attenuation are typically designed to ignore changes resulting from the temperature or the concentration of dissolved salts in the liquid (and thus changes in the speed of sound). The technique is also unaffected by flow or movement of the liquid. To a small degree, the acoustic properties of the carrier liquid will affect the measurement, so that particularly for low percentage solid applications, the zero solids adjustment should be made in the appropriate mother liquor. The major factor responsible for erroneous readings in most applications is another type of suspended scatterer, such as small gas bubbles, or a reflector in the beam such as a large solid (e.g. a stone) or large air bubbles. The installation must be made remembering that the effect of air bubbles is severe. This means that the technique is not applicable to monitoring biologically active liquids, for example fermenting liquids or digested sewage, or liquids where gas bubbles are produced, as in cooking processes for marmalade or jams. Where the suspended solids themselves contain, or can trap, variable

amounts of air, the slurry must be monitored with care, and not allowed to free-fall or be subject to violent agitation with possible air entrainment before it is monitored.

Liquid identification and interface detection systems

Ultrasonic attenuation measurements can, in theory, be used to identify the difference between two liquids by the different signal attenuation observed on placing a given ultrasonic sensor in the two liquids. This effect is a result of the different acoustic impedances (functions of bulk modulus and density) of the two liquids. These determine the proportion of the signal energy transmitted from the stainless steel transducer face into either of the liquids. Practical applications for this type of detection exist in distinguishing between crude oil or diesel fuel and seawater, and this is used extensively in offshore storage systems, for example in caissons in the Persian Gulf. Only when the two liquids are significantly different, as in this case of oil and water, can this technique work reliably, and no food industry applications in current use are known.

If, however, the interface between two liquids itself is used to cause a discontinuity, which interrupts or diverts the ultrasonic beam, then a significant signal loss between transmitter and receiver can occur, which is the basis of a practical system. This is frequently used for interface detection on immiscible liquids where the sensor is angled at approximately 15° to the interface (Fig. 12.15). If the interface lies across the beam, the ultrasonic signal is either reflected or refracted away from the receiver; no signal is detected. This is used in liquid/liquid separation systems to monitor interfaces, for example in the extraction of flavours and fragrances, or to control the thickness of acetone or similar organic liquids in biochemical culturing vessels.

In an emulsion produced by agitation or mixing of two immiscible liquids, the interfaces between these two liquids can affect an ultrasonic beam significantly, depending on the volume fraction and the size of second-phase globules, and on frequency (see Section 13.5.3). As a result, the ultrasonic transmission can be highly attenuated. This is used for detection systems in the petrochemical industry, giving an alarm output indicating the presence of an oil/water emulsion in the discharge line from a separator, for example. Such systems are less applicable in the food industry, except perhaps in the dairy sector, where the effect is potentially usable for the detection of cream layers and suspended inhomogeneously distributed fat inclusions.

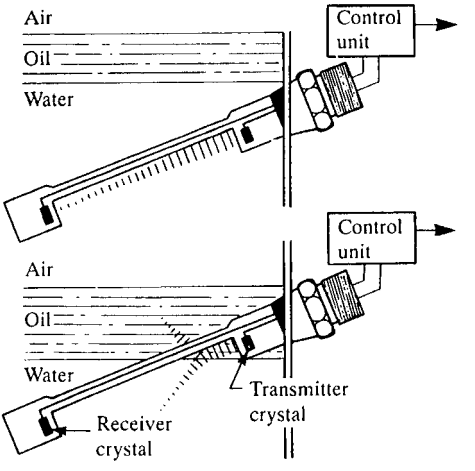


Fig. 12.15 Interface detection system by reflection.

12.3.5 Concentration measurement systems

A pulse of ultrasound travels through a liquid at a velocity dependent on the physical properties of the liquid, namely the bulk modulus of compressibility and the density. These are not typically the parameters used as inputs to a process control system. However, they vary with concentration changes of dissolved salts, or other solutes, and as a function of temperature. With the availability of microprocessor intelligence, the sometimes complex changes with temperature can be stored in look-up tables in the sensor electronics, and a sensor created that corrects for these temperature effects. The measurement displayed can be expressed in concentration, or other desired parameter, for a two-part liquid mixture.

A sensor for measurement of the speed of sound in a liquid (Fig. 12.16) is typically a flange-mounted unit fitted with an ultrasonic transducer and a reflector. The transducer acts as both a transmitter and a receiver monitoring the pulse echo from the reflector, allowing the speed of sound to be monitored over a path length of 100–200 mm total. An alternative approach uses a clamp-on sensor, fixed to the outside wall of a pipe, or tank. The pulse transmitted through the wall reflects from the opposite side of the pipe or tank, across the liquid. This is presumably a known path length, and so the installation can be calibrated.

The major UK Food industry application in liquid monitoring has been for measurement of alcohol strength in distilleries and spirit processing, to Customs and Excise requirements (Canongate 2000). This system uses a sensor operating across the full width of a storage tank, which is also used to feed speed of sound data into the pulse echo level measurement system working on the same tank (through the liquid up from the bottom, see Fig. 12.12)

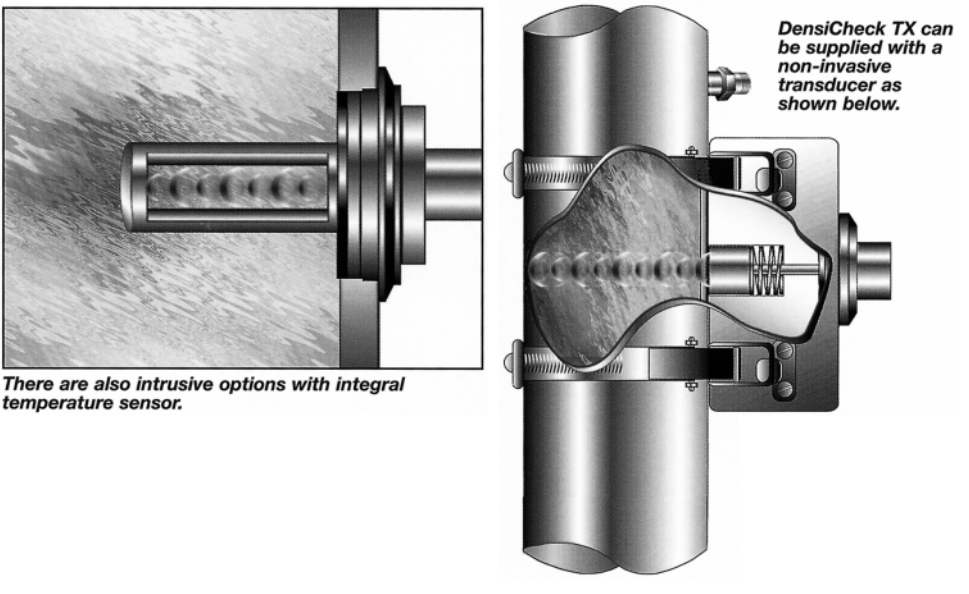


Fig. 12.16 (a) Concentration measurement using speed of sound measurement; complete wall-mounted transducer with built-in reflector. (b) Clamp-on concentration monitor using speed of sound measurement, working across a process pipe.

Ultrasonic measurement using a probe inserted into a pipeline is used to measure concentration of flowing fruit juice, over the range 0–25° BRIX (Mesa Laboratories 2000). The dilution level used to create consumable juice is critical, to both taste and profitability. The typical relationship between speed of sound and ° BRIX is shown in Fig. 12.17. Other similar applications are quoted in blending of oils, molasses, syrups etc., plus monitoring solutions of salt, sodium hydroxide, acids.

The major limitation of this technique is that the solute/solvent materials must be known and that the characteristic curves must be available from suppliers' test data, or these curves must be experimentally established for the solvent-solute system involved. In general the suppliers of equipment undertake this initial investigation work, if the relationship is not already known. The results will then apply to that solution provided that no other solutes or solvents are present in variable amounts, since these would produce their own variations in the speed of sound. The major application is therefore in simple mixtures, that is one solute and one solvent, preferably over a limited temperature range.

12.3.6 Future developments in high-frequency instrumentation

In high-frequency techniques, the main future developments will relate to the use of clamp-on transducers because of the hygiene advantage this offers. The variable nature of the bond between transducer, pipe or vessel wall, and liquid gives rise to signal amplitude variations. Techniques that use a time measurement will therefore be easier to apply than those that rely on an amplitude measurement. Already, the Doppler flow meter (which measures a frequency shift) is available as a clamp-on system. Time-of-flight ultrasonic flow meters will be further developed as clamp-on devices, probably with microprocessor-based electronics that allow a read-out of flow velocity and pipe wall thickness (to allow flow volume computation).

Already some Doppler type flowmeters that use advanced signal processing techniques are quoted in the literature (Panametrics 2000). Cross-correlation techniques, and pulse coded modulation of the transmitted ultrasonic signal will be developed. This

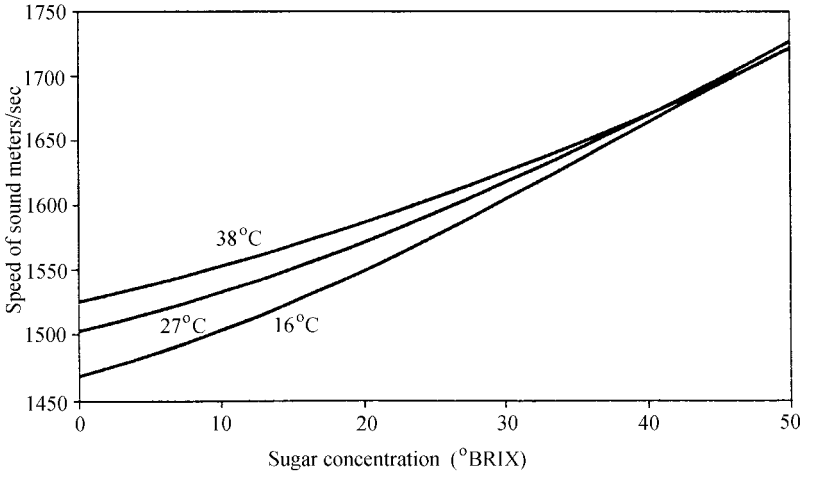


Fig. 12.17 Sound velocity versus sugar concentration in fruit juice (°BRIX) (Courtesy of NuSonics Divn, Mesa Laboratories Inc).

will allow time gating of the received signal, so that the Doppler measurement can be made on reflections from the pipe centre-line; this is an extension of the medical work on blood flow monitoring. Alternative methods for processing the scattered Doppler signals include speckle tracking or stroboscopic scattering (processing in the time domain) (Brown and Lynnworth 2001). A novel method of using a dual-frequency transmitter has recently been introduced (Polysonics 2000). Their DFD system is claimed to reduce noise effects from the plant and flow, by frequency shifting the second received waveform and correlating or multiplying the two together. Because the noise will have the same frequency in each received spectrum, which is then shifted, the product of the two noise waveforms will be reduced, whereas the Doppler signal is accentuated.

Developments in the use of microprocessor intelligence available will allow multivariable measurement techniques, which will measure the speed of sound in the liquid, and use temperature inputs to deduce the composition, plus calibrate the built-in time-of-flight flow cell using this speed of sound where necessary, to give flow rate. Ultrasonic signal attenuation measurement systems offer several potentially useful applications in the food industry, for which the basic research work and application engineering necessary are being undertaken by various research laboratories. Through tripartite co-operation between these laboratories, instrument suppliers and process plant users, useful and viable developments could arise. However, some of the techniques described here have been developed originally for use in other industries, and then have found spin-off applications in the food industry after only slight modification. The ultrasonic point level sensor is an example of this, and developments can be expected in single-sided clamp-on level switches or special hygienic sections of pipe-work incorporating such sensors.

12.4 Contacts for further information

The science of ultrasound propagation applied to measurement or sensing is fascinating in any industry, but is particularly relevant to the food industry with the potential advantages described above. For practical applications of the standard techniques, the appropriate manufacturers can provide literature and give advice based on current experience (Solartron Mobrey 1975–91; Canongate 2000; Panametrics 2000; Pulsar 2000; Flotec/Valeport 1999; Mesa Laboratories 2000). For research into specific problems and for advice on the more difficult industrial applications there are several centres of ultrasonic expertise. For the UK, these are associated with the nuclear industry (for example, Instrumentation and Applied Physics at Harwell, and the Northern Research Laboratories at Risley) or are based in academic institutions (Department of Fluid Engineering and Instrumentation, Cranfield, and departments at Bradford and Leeds Universities and Teesside Polytechnic). Food industry applications in the UK have been pioneered by groups working at the Leatherhead Food Research Association, the Brewing Research Foundation, the Institute of Food Research Laboratories (both at Norwich and Langford near Bristol). The work of the Langford group involved with ultrasonic measurements on meat has now moved to the University of Bristol (C. A. Miles). The Campden Food and Drink Research Association is also involved with ultrasonic studies.

Ultrasonics today offers a significant range of useful instrumentation to the food industry. The standard products have been described, with information on their limitations and likely developments. More specialised products are also available, but

need care in their applications at present. Once their valid applications are established by practical testing, these will become the hygienic, non-contact, non-penetration sensors for many of the required process parameters of tomorrow. Ultrasonic techniques are growing in importance to the food process engineer and can give significant advantages in this industry.

12.5 Acknowledgements

The experience related in this chapter represents the result of working in this industry for many years, primarily for Solartron Mobrey of Slough, and more recently on clamp-on flowmeter developments with Platon and other researchers. Without the help and assistance of colleagues, and the customers, none of this would have been possible.

The assistance and suggestions made by Larry Lynnworth of Panametrics in the preparation of this chapter are gratefully acknowledged. The research into transducers, particularly for flow measurement systems, and the willingness to share ideas demonstrated by Larry Lynnworth and his team at Panametrics over the years, is an example to be praised.

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